

- 2 An object of mass 1.5 kg is released from a stationary hot air balloon. Fig. 2.1 shows how the vertical displacement of the object varies with time.

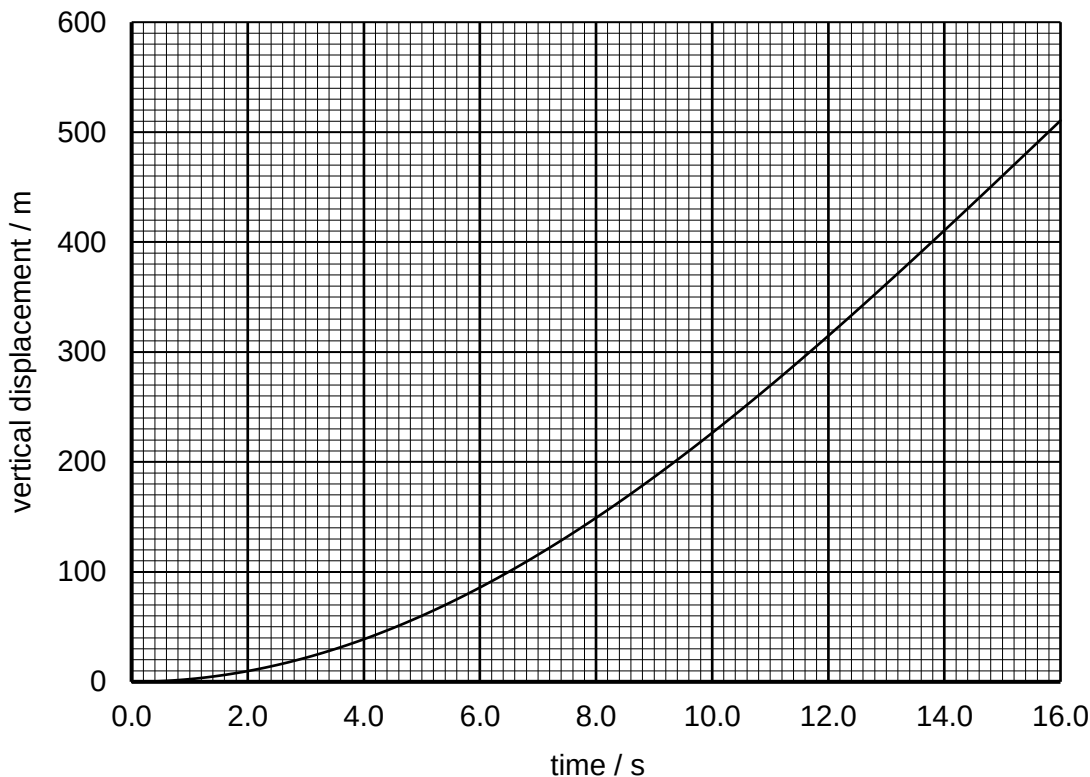


Fig. 2.1

- (a) Calculate the change in gravitational potential energy ΔE_p of the object that occurred during the 16 s after it was released.

$$\begin{aligned}
 \Delta E_p &= mgh \\
 &= 1.5 (9.81)(-510) \\
 &= (-) 7500 \text{ J } \textcolor{red}{[A1]}
 \end{aligned}$$

$$\Delta E_p = \dots\dots\dots \text{ J } [1]$$

- (b) Using Fig. 2.1, determine the speed of the object at $t = 16$ s.

$$\begin{aligned}\text{speed} &= \text{gradient of tangent to the graph at } t = 16 \text{ s} \quad [\text{M1}] \\ &= 50 \text{ m s}^{-1} \text{ (acceptable range: } 49 \text{ m s}^{-1} \text{ to } 52 \text{ m s}^{-1}) \quad [\text{A1}]\end{aligned}$$

$$\text{speed} = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- (c) Calculate the change in kinetic energy ΔE_k of the object during the same period.

$$\begin{aligned}\Delta E_k &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \\ &= \frac{1}{2}(1.5)(50)^2 - 0 \\ &= 1880 \text{ J} \quad [\text{A1}]\end{aligned}$$

$$\Delta E_k = \dots\dots\dots \text{ J} \quad [1]$$

- (d) Explain why ΔE_p and ΔE_k are not equal to one another.

Work is done (by object) against/to overcome air resistance. [B1]

.....
..... [1]

- (e) The object strikes the ground 16 s after it was released and penetrates 0.85 m into the ground. Determine the average resistive force acting on the object as it penetrates the ground.

Work done against resistive force = Loss of E_k (and E_p)

$$F(0.85) = 1880 \quad [\text{M1}] \quad (+ 1.5 \times 9.81 \times 0.85 \quad [\text{B1}])$$

$$F = 2210 \text{ N (or } 2230 \text{ N)} \quad [\text{A1}]$$

$$f - mg = ma$$

$$\left(-1.5(9.81) \right)^{[\text{B1}]} + f = 1.5 \left(\frac{50.0^2}{2 \times 0.85} \right)^{[\text{M1}]} \quad \text{average resistive force} = \dots\dots\dots \text{ N} \quad [3]$$

$$= 2210 \text{ N (or } 2230 \text{ N)} \quad [\text{A1}]$$

[Total: 8]

2 Fig. 2.1 shows a 0.45 kg mass held on a horizontal air track by two identical springs that are